Stochastic constrained factorization for low-dose CT

2DeteCT pilot and SFRAMES initialization ablation

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Abstract

We compare SFRAMES, SPRING–SARAH, ProxSARAH, and PALM on a common nonconvex, rank-64 reconstruction model for measured 2DeteCT data. The evaluation covers two slices and two training-view counts per slice. SFRAMES leads the shared-start comparison at the reported time and work budgets. A paired initialization experiment starts SFRAMES on its Frobenius caps and outside the nonnegative orthant, then tracks feasibility and loss under the original optimizer settings.

1 Model and stochastic oracle

The unknown attenuation image is $X = UV^\top$ on a 512×512 grid. Both factors lie in $\mathbb{R}^{512\times 64}$. All angular views of a slice constrain the same factors, so the number of measurements can vary at fixed image size and rank. In the SFRAMES template, the variable is $x = (U, V)$ and the problem is

$$\min_{x\in C} f(x) + g(Tx), \tag{1}$$

$$C = \{(U,V) :\ \|U\|_F \le R_U,\ \|V\|_F \le R_V\}, \tag{2}$$

$$D = \{(U,V) :\ U \ge 0,\ V \ge 0\}, \qquad g = \iota_D, \qquad T = \mathrm{Id}. \tag{3}$$

The inequalities are entrywise. The indicator ι_D is zero on D and $+\infty$ elsewhere. The identity T acts on the factors. The CT forward operator acts on their product inside f.

What the scanner measures. An attenuation image describes how strongly the material at each location reduces the transmitted X-ray beam per unit distance. At one angular position, called a *view*, an X-ray source illuminates a line of detector bins on the opposite side of the object. The rays from the source to those bins spread out in a fan. Each bin records the intensity transmitted along its ray. In the idealized Beer–Lambert model, dividing the transmitted intensity by the unobstructed intensity and taking the negative logarithm gives the line integral of attenuation along that ray. This is why the processed measurements can be fitted with a linear forward operator in the image X [2].

The forward operator P_i. Treat X_{pq} as the constant attenuation within pixel (p, q). Let $\ell_{ij,pq}$ be the length in millimeters of ray j in view i inside that pixel. The CPU line projector used here applies the map

$$P_i \colon \mathbb{R}^{512\times 512} \longrightarrow \mathbb{R}^d, \qquad [P_i(X)]_j = \sum_{p,q} \ell_{ij,pq} X_{pq}, \qquad d = 956. \tag{4}$$

The weights are fixed by the calibrated source and detector geometry. A ray contributes only to pixels it crosses. ASTRA traces the rays to apply the operator, so the solver can evaluate a selected set of views directly [4].

Write $\mathcal{I}_{\text{train}}$ for the N training view indices and $y_i \in \mathbb{R}^d$ for the processed low-dose measurements at view i. The unweighted least-squares objective is

$$f(U,V) = \frac{1}{N} \sum_{i \in \mathcal{I}_{\text{train}}} q_i(U,V), \qquad q_i(U,V) = \frac{\|P_i(UV^\top) - y_i\|_2^2}{2d}. \tag{5}$$

Each view contributes all its detector bins. The image-space loss is quadratic and the factorized objective is quartic and nonconvex. The component gradients are Lipschitz on the compact feasible region C. Global smoothness on the ambient factor space would require an extension outside a neighborhood of C. Small strictly positive factors lie in the interiors of both C and D.

Backprojection and stochastic gradients. The adjoint P_i^*, called *backprojection*, distributes each detector residual back to the pixels crossed by its ray, using the same weights $\ell_{ij,pq}$. Thus $P_i^*(P_i(X) - y_i)$ is the image-space gradient of the unnormalized half squared residual. The chain rule through $X = UV^\top$ then gives the two factor gradients.

At each SFRAMES update, draw a fresh uniform 64-element subset $B \subset \mathcal{I}_{\text{train}}$ without replacement within the batch and independently between updates. The gradient estimate is

$$E_B = \frac{1}{|B|d} \sum_{i \in B} P_i^*(P_i(UV^\top) - y_i), \qquad \widehat{\nabla f}_B(U,V) = (E_B V, E_B^\top U). \tag{6}$$

This estimate is unbiased conditional on the current iterate. One joint gradient evaluation uses one forward projection and one adjoint restricted to B.

2 Data and acquisition geometry

The inputs are measured fan-beam sinograms from 2DeteCT [2]. Mode 1 supplies low-dose observations for fitting and held-out-view diagnostics. The image reference for each slice is the published Mode 2 nonnegative least-squares reconstruction from a separate high-dose acquisition. The development slice is `04300`. The evaluation slices are `04531` and `04800`. The official Zenodo archives supplied slices `04300` and `04800`, and the DeepInverse single-slice mirror supplied `04531` [3]. Experiment data sidecars `data/slice*.json` record download sources and TIFF checksums.

A *sinogram* stacks the detector readings from all angular views. Each raw Mode 1 sinogram has 3,601 views and 1,912 detector cells. Discard the repeated last view and sum adjacent detector cells. Apply the same binning to the dark field, which measures the detector offset, and the mean of the two flat fields, which measure unobstructed illumination. If $S_i, D_{\text{dark}}, F_{\text{flat}}$ denote the resulting arrays, the pre-reversal measurement is

$$-\log\left(\max\left\{\frac{S_i - D_{\text{dark}}}{F_{\text{flat}} - D_{\text{dark}}}, 10^{-6}\right\}\right).$$

Reverse the detector axis to match the DeepInverse convention [3]. The recorded clipping count was zero for each of the three sinograms. Each processed acquisition has $3{,}600 \times 956$ entries.

We use the ASTRA 2.5.0 CPU `line_fanflat` projector in physical millimeters. The source-to-origin distance is 431.019989 mm, the origin-to-detector distance is 97.980499 mm, and the binned detector spacing is 0.1496 mm. The square image field has width 116.528143 mm and center $(-0.113797, -0.113797)$ mm. The angle of view i is $\pi - 2\pi i/3600$. Vertically flip each published 1024×1024 reference, divide by the native voxel width 0.113797 mm, and average 2×2 blocks to align with the 512×512 fitting grid.

Development runs use the smaller training set. Both training sizes are evaluated on each evaluation slice. Calibration views are a subset of both training sets and are disjoint from validation. The training split fixes which measured views define the objective. The optimizer then samples minibatches from

Table 1: Exact view splits. Indices i range from 0 to 3599.

Role	Views	Selection rule
Smaller training set	360	$i \bmod 10 = 0$
Larger training set	1,080	$i \bmod 10 \in \{0, 3, 6\}$
Held-out validation	360	$i \bmod 10 = 9$
Calibration	120	$i \bmod 30 = 0$

that fixed split. Held-out loss measures how well the reconstruction predicts separate low-dose views of the same slice, while reference-image error compares the reconstructed attenuation values with the high-dose reconstruction.

3 Initialization and optimizer settings

Feasible calibration. Let P_J stack the 120 calibration views. Divide the positive backprojection $(P_J^* y_J)_+$ entrywise by $P_J^* P_J \mathbf{1}$, floored at 10^{-3} of its maximum. Initialize rank-64 nonnegative factors with an NNDSVD-style factorization of this image. Add 10^{-3} times each factor's RMS times independent uniform $[0, 1)$ entries. Fit a scalar amplitude by least squares on the calibration views with lower bound 10^{-6}, then multiply each factor by its square root. Denote the resulting factors by $U_{\mathrm{cal}}, V_{\mathrm{cal}}$. Each Frobenius cap is twice the norm of the corresponding calibrated factor.

Estimate λ by ten power steps on $P_J^* P_J$, starting from a normalized all-ones image, and divide the resulting eigenvalue estimate by $120d$. The empirical block curvature scales are

$$L_U = \lambda \|V_{\mathrm{cal}}\|_{\mathrm{op}}^2, \qquad L_V = \lambda \|U_{\mathrm{cal}}\|_{\mathrm{op}}^2.$$

Both training sizes use the same calibration and caps within each slice. Table 2 gives the resulting values.

Table 2: Per-slice Frobenius caps, initial curvature scales, and smoothing prefactor.

Slice	R_U	R_V	L_U	L_V	μ_0
04300	4.12869674	4.12867862	0.19331899	0.19443450	5.14312016
04531	3.81305622	3.81304219	0.16367101	0.16325009	6.10981743
04800	4.22469276	4.22465958	0.20220242	0.20283452	4.93012733

SFRAMES. For $k \geq 0$, initialize momentum with the first sampled gradient and then use

$$m_k = (1 - \rho_k) m_{k-1} + \rho_k \widehat{\nabla f}_{B_k}(x_k), \qquad \gamma_k = \frac{\gamma_0}{(k+1)^{3/4}}, \qquad \rho_k = \frac{1}{(k+1)^{1/2}}, \qquad \mu_k = \frac{\mu_0}{(k+1)^{1/4}}.$$

The momentum formula applies for $k \geq 1$, and $\mu_0 = 1/\max(L_U, L_V)$. The exponents follow the SFRAMES manuscript [1]. Add the Moreau gradient after the momentum update and take the Frank–Wolfe step

$$s_k = \mathrm{LMO}_C\left(m_k + \frac{x_k - \Pi_D(x_k)}{\mu_k}\right), \qquad x_{k+1} = (1 - \gamma_k) x_k + \gamma_k s_k.$$

The projection Π_D clips negative entries. For a block direction $G \neq 0$, the LMO on the radius-R Frobenius ball is $-RG/\|G\|_F$. At $G = 0$ it returns zero.

Paired LMO start. The original four-way comparison uses $U_{\mathrm{cal}}, V_{\mathrm{cal}}$ for all methods. The SFRAMES ablation retains the caps, curvature scales, rate settings, view splits, and work budgets. It starts with one LMO atom for each factor using independent standard Gaussian block directions. This places the start on the Frobenius caps and outside D, and replaces the data-informed initial directions. Both variants start with empty momentum. The base seed is 110 in development and 201 in evaluation. Minibatches use NumPy's separate generator seeded with the base plus 100,000. The LMO directions use the base plus 200,000. Within each original–rerun pair the minibatch stream is identical. The two evaluation view counts share the same LMO factors within each slice.

Comparators. SPRING-SARAH updates U and then V with the new U, using independent full refresh probability 1/16 for each block and 64-view recursive differences between refreshes. Both blocks begin with a full gradient at their respective query points [5]. ProxSARAH uses the joint factor gradient, with 256-view snapshots, 12 recursive updates of 64 views per snapshot, and relaxation 0.8 [6]. Each recursive difference evaluates the same sampled views at its two query points. PALM uses the full training set at each alternating block step [7]. Their projection onto $C \cap D$ takes the positive part and radially clips each block to its cap if necessary.

Table 3: Development grids and frozen evaluation step rules. The rate multiplier is a.

Method	Candidate rates	Selected	Step rule
SFRAMES	0.10, 0.25	0.25	$\gamma_0 = a,\ \rho_0 = 1$
SPRING-SARAH	0.30, 0.70	0.70	$(\eta_U, \eta_V) = (a/L_U, a/L_V)$
ProxSARAH	0.40, 0.90	0.90	$\eta = a/(L_U + L_V)$
PALM	0.60, 1.10	1.10	$(\eta_U, \eta_V) = (a/L_U, a/L_V)$

The selected rate minimizes repaired training loss at the last development probe within 5,760 forward views. The table reports the practical pilot settings. The initialization ablation repeats both SFRAMES development rates and the frozen evaluation rate without selecting a new rate.

4 Work accounting and diagnostics

Each run stops after the first completed update that reaches its target of 5,760 forward views in development or 21,600 in evaluation. Save probes at iteration zero, after the first update, upon crossing multiples of 720 or 2,160 views respectively, and at the endpoint. SFRAMES completes 90 development updates at 5,760 views and 338 evaluation updates at 21,632 views. Forward and backprojected views are recorded separately. Counts include refreshes and both gradient evaluations in a SARAH difference.

Raw training loss is f at the actual iterate. Repaired training loss and held-out loss use the projection onto $C \cap D$. The latter uses the normalization in (5) over the held-out set. Relative reference-image error is the Frobenius error of the repaired reconstruction divided by the norm of the aligned Mode 2 image. For the raw factors, the distance to D is the product Frobenius norm of their negative parts. Maximum violation is the maximum of zero, the two cap excesses, and the largest negative-entry magnitude in either block. Repair and diagnostic evaluations leave the optimizer state unchanged.

Elite used FP32 ASTRA forward and adjoint operations and FP64 factor arithmetic. Each worker used one BLAS thread on a distinct physical core. Optimizer timings include projections and oracle calls and exclude calibration, diagnostic probes, and checkpoint I/O. Times reflect concurrent arms on the same host.

5 Results

5.1 Four methods with a common feasible start

All eight development runs and all sixteen evaluation runs completed. Each evaluation case used a common initialization for the four methods. We compare the last saved probe within each budget, with no interpolation. At both 100 optimizer seconds and 21,600 forward-projected views plus 21,600 backprojected views, SFRAMES had the lowest repaired training loss, held-out view loss, and relative reference-image error in every evaluation case. Table 4 gives the medians across the four slice and view-count combinations at the projection budget. Appendix A gives the curves for each evaluation case.

Table 4: Median at the last saved probe within 21,600 forward views and the same number of backprojected views. Held-out loss and relative error against the high-dose reference use the repaired factors.

Method	Raw training	Repaired training	Held-out	Reference error
SFRAMES	0.015717	0.015724	0.015768	45.17%
ProxSARAH	0.022315	0.022315	0.022347	55.81%
SPRING-SARAH	0.024481	0.024481	0.024504	59.12%
PALM	0.027301	0.027301	0.027329	62.59%

At the 100-second budget the median repaired training losses were 0.01586, 0.02327, 0.02523, and 0.02780 in the same order. All evaluation arms reached their stopping budget after approximately 110–114 optimizer seconds. At the matched projection budget, repair increased the SFRAMES median training loss by about 0.04%. The other three methods were feasible to numerical precision at those probes.

5.2 Paired SFRAMES initialization ablation

The Gaussian-direction LMO starts were on the Frobenius caps and outside the nonnegative orthant. Their initial distances to D were 4.1583 on development slice 04300 and 3.8221 and 4.2347 on evaluation slices 04531 and 04800. Each original start had zero distance to D. All six reruns matched the original view counts at every saved probe and matched the terminal minibatch RNG state. The largest observed cap excess in the reruns was 8.9×10^{-16}.

The development pairs ended after 90 updates and 5,760 views in each projection direction. The evaluation pairs ended after 338 updates and 21,632 views in each direction. Table 5 compares the paired endpoints.

Table 5: Paired SFRAMES endpoints for the original warm start and the Gaussian-direction LMO start. Distance to D is measured on the raw factors.

			Repaired training loss		Distance to D	
Slice	Views	γ_0	Warm	LMO	Warm	LMO
04300	360	0.10	0.019652	0.036069	0.007777	0.001284
04300	360	0.25	0.018151	0.018474	0.008689	0.006256
04531	360	0.25	0.013276	0.012975	0.004310	0.003932
04531	1,080	0.25	0.013294	0.012999	0.003984	0.003890
04800	360	0.25	0.017926	0.017668	0.003934	0.003814
04800	1,080	0.25	0.017979	0.017706	0.003846	0.003683

At the matched evaluation endpoints, the LMO start reduced repaired training loss by 1.4–2.3% in all four pairs. Median held-out loss fell from 0.015658 to 0.015378, and median relative reference-image error fell from 44.88% to 44.12%. Both shorter development pairs finished with higher repaired loss from the LMO start, especially at $\gamma_0 = 0.10$. The LMO start replaces the data-informed factors with random signed atoms, so the differences reflect both initial infeasibility and the change in starting information.

Scope. The evaluation spans two slices, two training-view counts per slice, and one evaluation seed. The larger candidate rate won for every method on the single development slice. The ordering therefore describes the tested rates and work budgets. The paired initialization results compare SFRAMES starts, while the four-method results use the original common warm start.

References

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[2] M. B. Kiss et al. *2DeteCT: A large 2D expandable, trainable, experimental Computed Tomography dataset for machine learning*. Scientific Data, 10, article 576, 2023. DOI.

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[4] ASTRA Toolbox. *2D Projectors*. Fan-beam line projector documentation.

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[6] N. H. Pham, L. M. Nguyen, D. T. Phan, and Q. Tran-Dinh. *ProxSARAH: An Efficient Algorithmic Framework for Stochastic Composite Nonconvex Optimization*. Journal of Machine Learning Research, 21(110), 1–48, 2020. Article.

[7] J. Bolte, S. Sabach, and M. Teboulle. *Proximal alternating linearized minimization for nonconvex and nonsmooth problems*. Mathematical Programming, 146, 459–494, 2014. DOI.

A Four-method comparison plots

The evaluation arms in these figures share the data-informed feasible start.

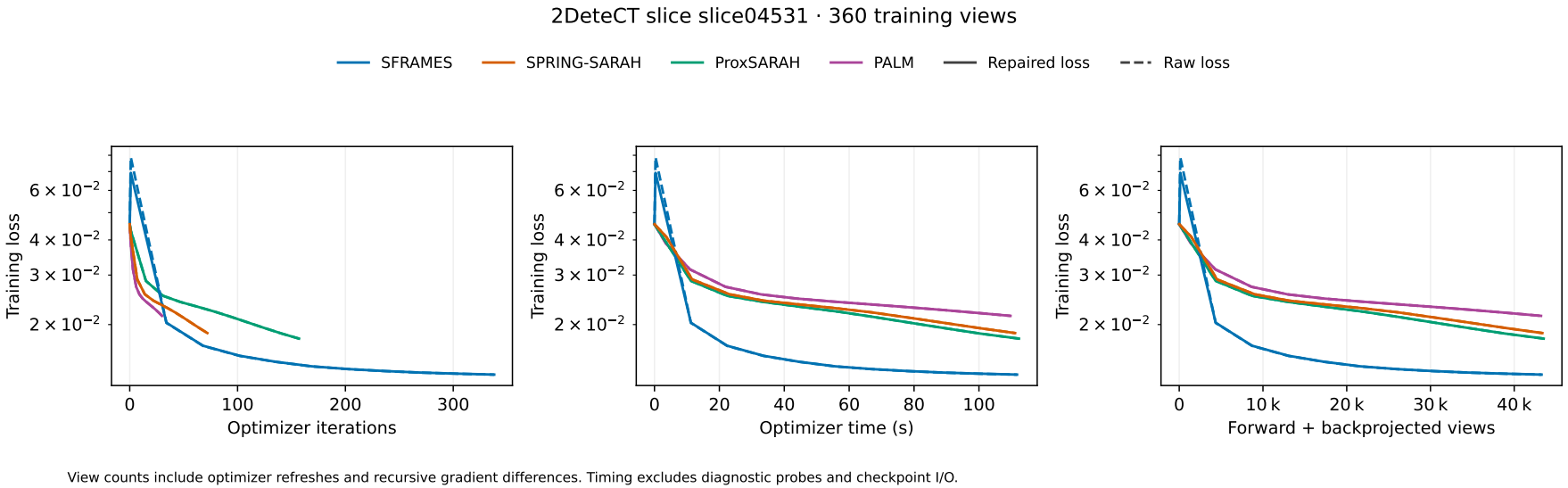


Figure 1: Slice 04531 with 360 training views. Raw and repaired training loss by update count, optimizer time, and projection work.

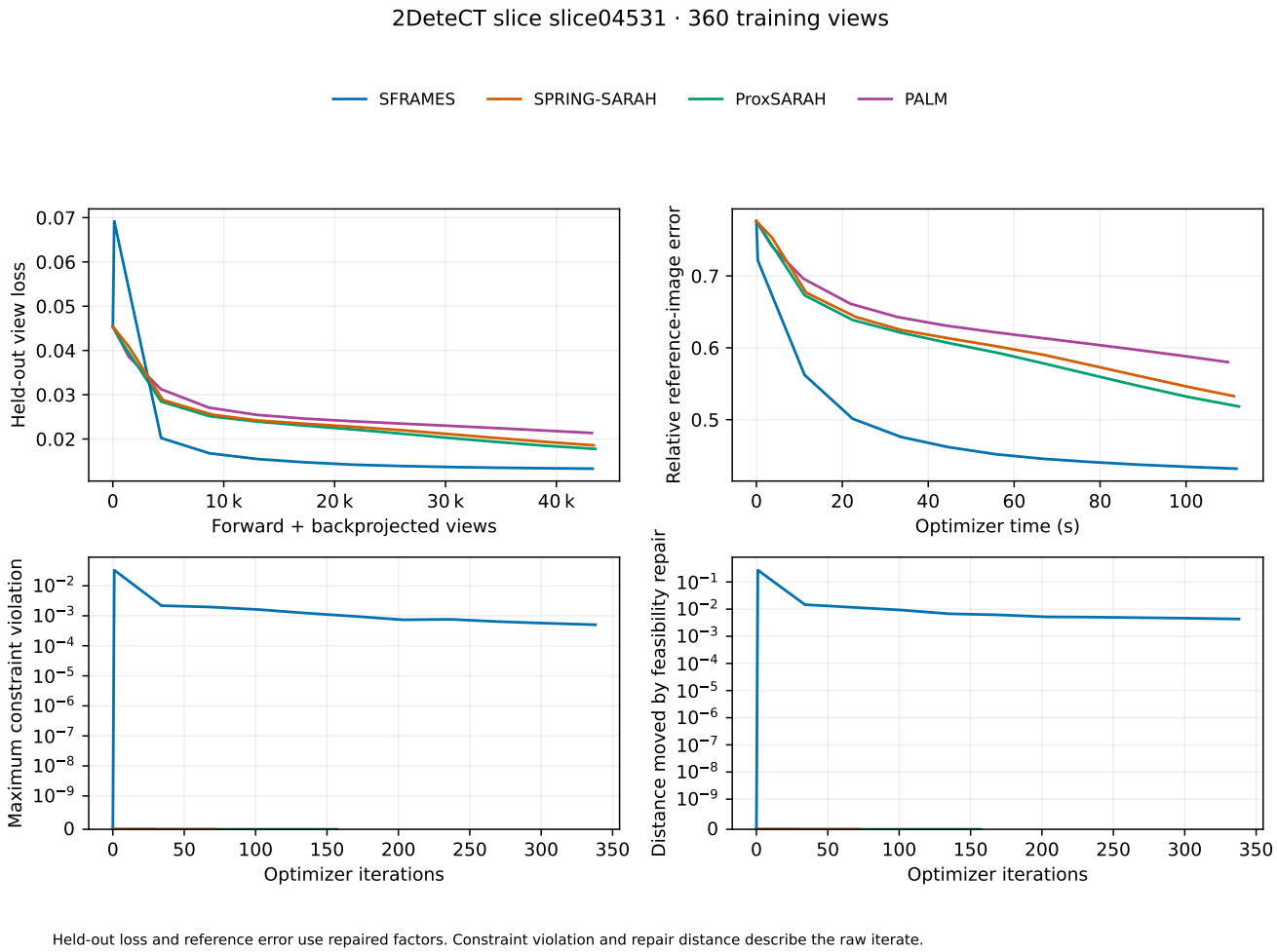


Figure 2: Slice 04531 with 360 training views. Held-out loss and relative reference-image error use repaired factors. Feasibility is measured on the raw factors.

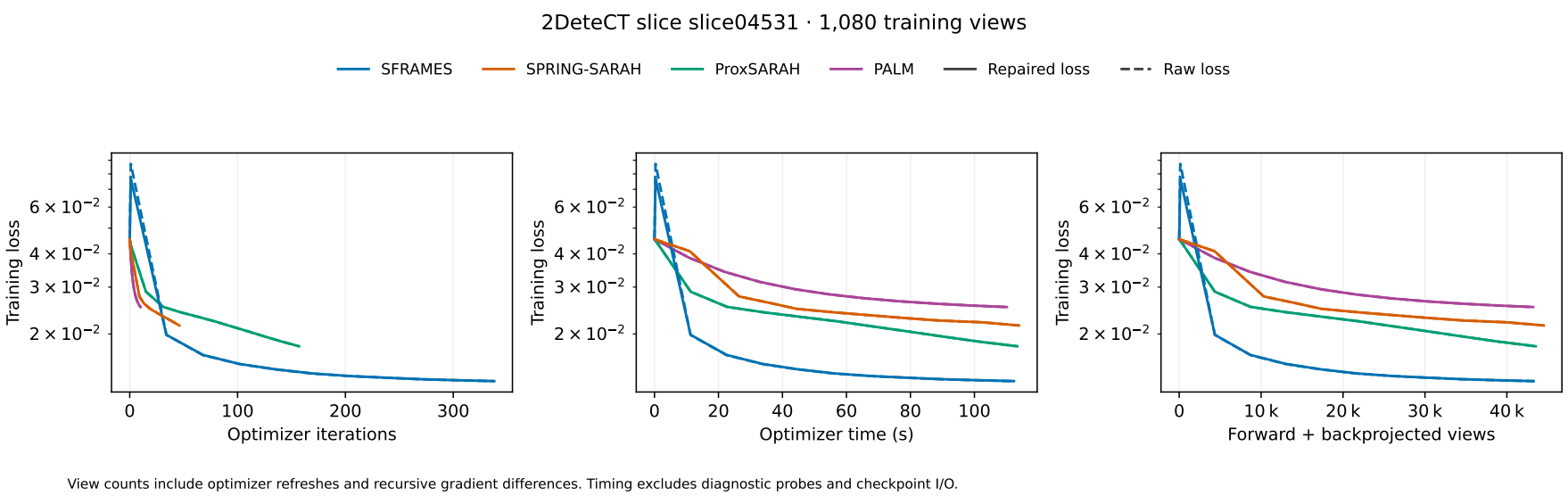


Figure 3: Slice 04531 with 1,080 training views. Training loss by update count, optimizer time, and projection work.

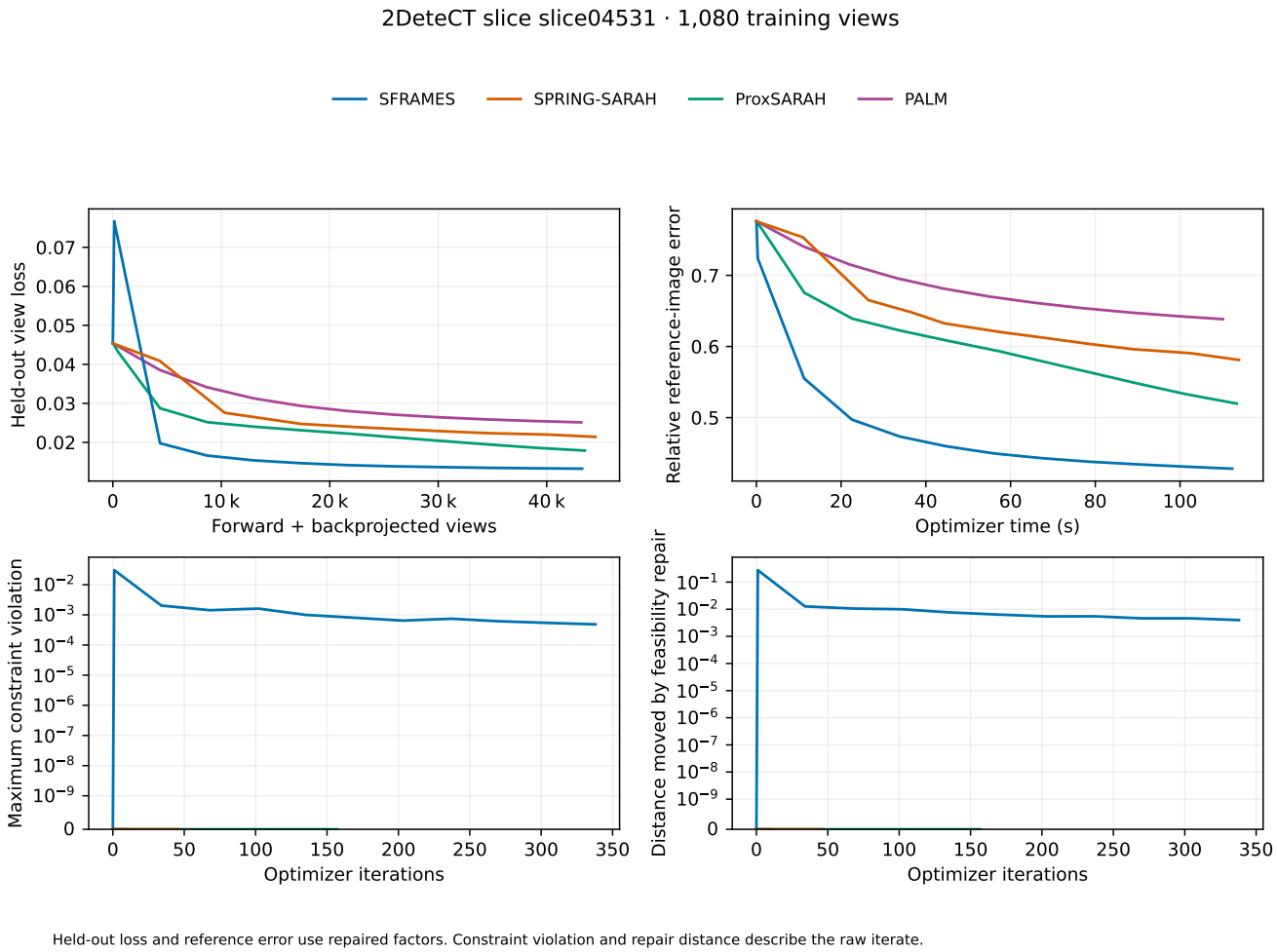


Figure 4: Slice 04531 with 1,080 training views. Held-out loss, reference-image error, and raw feasibility.

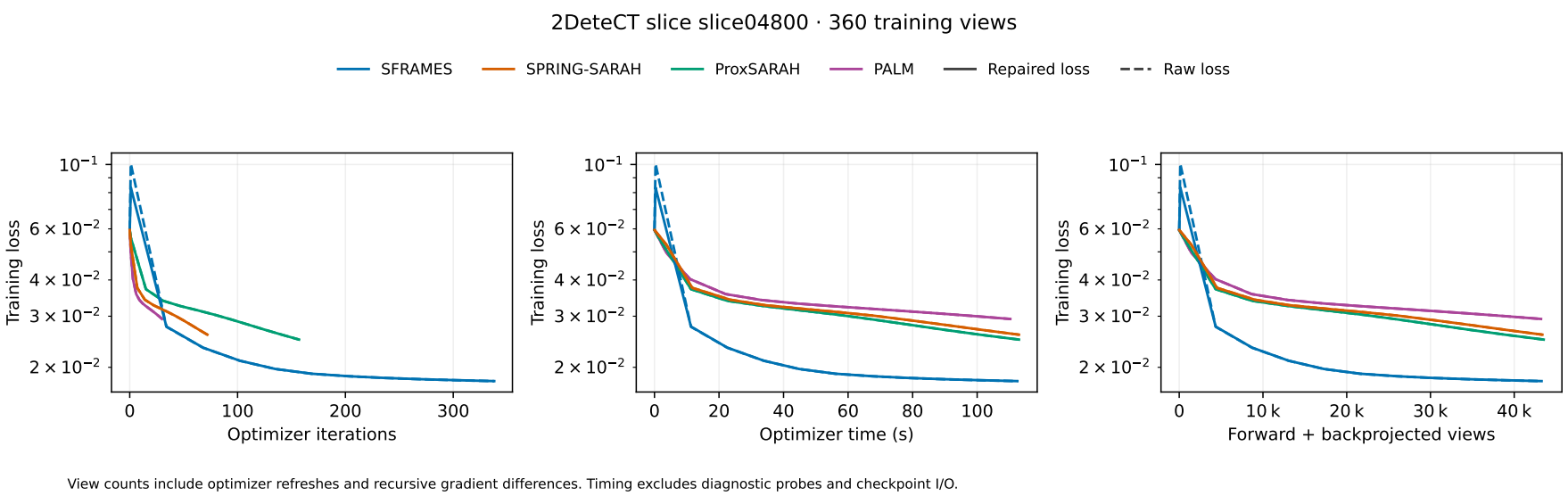


Figure 5: Slice 04800 with 360 training views. Training loss by update count, optimizer time, and projection work.

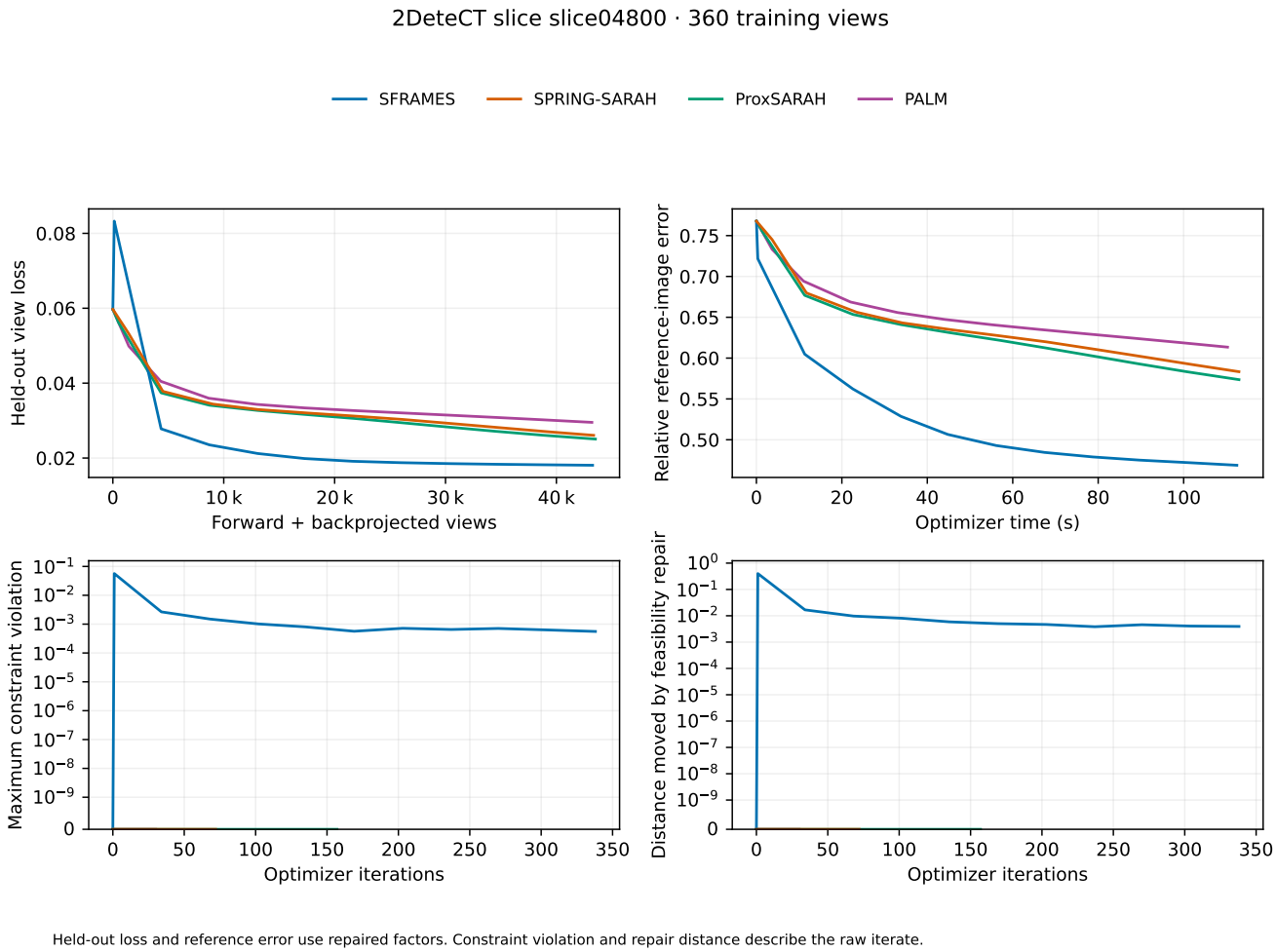


Figure 6: Slice 04800 with 360 training views. Held-out loss, reference-image error, and raw feasibility.

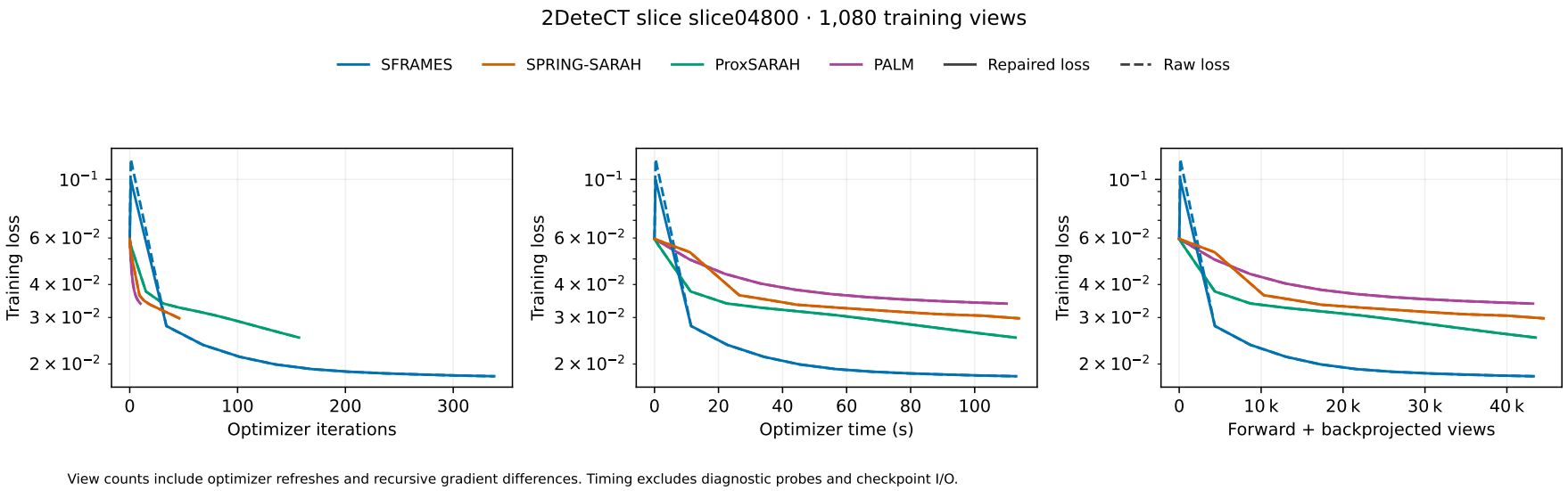


Figure 7: Slice 04800 with 1,080 training views. Training loss by update count, optimizer time, and projection work.

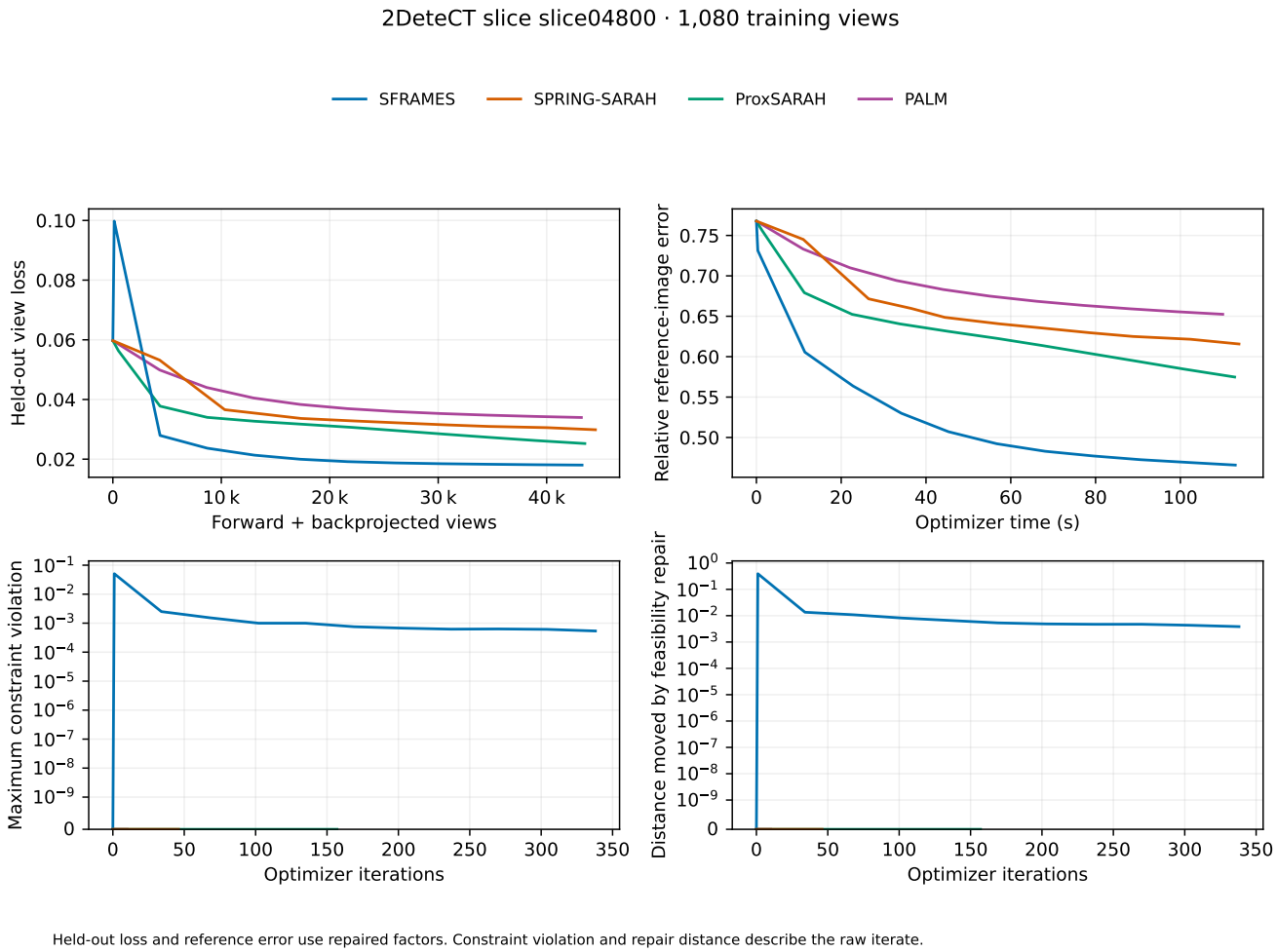


Figure 8: Slice 04800 with 1,080 training views. Held-out loss, reference-image error, and raw feasibility.